

Automated ACMG/AMP Variant Pathogenicity Classification for *STK11* (Peutz-Jeghers Syndrome): A Reproducible Bayesian Pipeline

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Abstract

Automated, standardized, and reproducible variant classification is critical for clinical genetics. Here, we present **stk11-peutz-pipeline**, an automated Snakemake workflow implementing the Tavtigian Bayesian point framework for the classification of variants in the *STK11* gene (associated with Peutz-Jeghers syndrome). The pipeline integrates variant normalization, VEP v115, SpliceAI, gnomAD v4.1.0, and ClinVar. Analyzing 1209 real ClinVar variants, the pipeline identified 168 clinically actionable (Pathogenic or Likely Pathogenic) variants, demonstrating 65.5% concordance with the commercial Franklin by Genoox platform.

Introduction

Peutz-Jeghers syndrome is a rare autosomal dominant disorder characterized by gastrointestinal hamartomatous polyps and an increased risk of various cancers, caused by mutations in the *STK11* gene [1]. Accurate classification of *STK11* variants according to the ACMG/AMP guidelines [2] is critical for patient surveillance and genetic counseling.

We present an automated, containerized pipeline that implements the Tavgian Bayesian point system [3] with ClinGen-recommended gene-specific calibrations for *STK11* variant interpretation.

Materials and Methods

The pipeline is implemented in Snakemake [4]. Variants are normalized with BCFtools [5], annotated with VEP v115 [6](#), and splicing effects are predicted with SpliceAI [9]. Population frequencies are retrieved from gnomAD v4.1.0 [10] and clinical records from ClinVar [11].

ACMG criteria are mapped to points: PVS1 (+8), PS (+4), PM (+2), PP (+1), BP (-1), BM (-2), BS (-4), BA (-8). The final classification thresholds are: Pathogenic (≥ 10), Likely Pathogenic (6-9), VUS (0-5), Likely Benign (-1 to -5), Benign (≤ -6).

Results and Discussion

We analyzed **1209 real ClinVar variants** in the *STK11* gene. The pipeline classified 126 as Pathogenic, 42 as Likely Pathogenic, 347 as VUS, 163 as Likely Benign, and 531 as Benign. A total of **168 variants (13.9%)** were classified as clinically actionable (Pathogenic or Likely Pathogenic).

Comparison with the commercial Franklin by Genoox platform showed **65.5% concordance**, with discordances primarily driven by conservative splicing predictions (SpliceAI) and in silico predictor (REVEL) thresholds in our pipeline.

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